Probability Density Functions

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1 Definitions

Definition 1 (Probability density function)**.** *Let X be a real-valued random variable on a probability space $(\Omega, \mathcal{F}, P)$. A non-negative, Lebesgue-integrable function $f_X : \mathbb{R} \to [0, \infty)$ is called the* probability density function *(PDF) of X if for every Borel set $B \subseteq \mathbb{R}$,*

$$P(X \in B) \;=\; \int_B f_X(x)\, dx.$$

Equivalently, the law of X is absolutely continuous with respect to Lebesgue measure, and $f_X = dP_X/d\lambda$ is the Radon–Nikodym derivative.

Definition 2 (CDF–PDF relation)**.** *The cumulative distribution function $F_X(x) = P(X \leq x)$ satisfies*

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt, \qquad f_X(x) = F_X'(x) \text{ a.e.}$$

Proposition 1 (Normalization)**.** *Any PDF f_X satisfies $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$.*

Proof. $\int_{\mathbb{R}} f_X = P(X \in \mathbb{R}) = 1$. □

2 The Gaussian PDF

Define the *standard Gaussian density*

$$\phi(x) \;=\; \frac{1}{\sqrt{2\pi}} \exp\!\left(-\frac{x^2}{2}\right).$$

Clearly $\phi > 0$, so to show ϕ is a PDF we need only verify normalization.

Theorem 1 (Gaussian normalization)**.** *The Gaussian PDF ϕ integrates to 1 over $\mathbb{R}$:*

$$\int_{-\infty}^{\infty} \phi(x)\, dx \;=\; 1.$$

Proof. Let $I := \int_{-\infty}^{\infty} e^{-x^2/2}\, dx$. The integral is finite by comparison with a step function (or by direct integrability check). We compute I^2 via Fubini's theorem on the product:

$$I^2 \;=\; \left(\int_{-\infty}^{\infty} e^{-x^2/2}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\, dy\right) \;=\; \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\, dx\, dy.$$

Switch to polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, with Jacobian r:

$$I^2 \;=\; \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2}\, r\, dr\, d\theta.$$

The inner integral evaluates by substitution $u = r^2/2$, $du = r\,dr$:

$$\int_0^\infty r\,e^{-r^2/2}\,dr \;=\; \int_0^\infty e^{-u}\,du \;=\; 1.$$

Therefore

$$I^2 \;=\; \int_0^{2\pi} 1\,d\theta \;=\; 2\pi,$$

so $I = \sqrt{2\pi}$ (taking the positive root, as $e^{-x^2/2} > 0$). Hence

$$\int_{-\infty}^{\infty} \phi(x)\,dx \;=\; \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} \;=\; 1. \qquad \square$$

Proposition 2 (Gaussian CDF)**.** *The standard Gaussian CDF* $\Phi(x) = \int_{-\infty}^{x} \phi(t)\,dt$ *admits no elementary closed form, but*

$$\Phi(x) \;=\; \tfrac{1}{2}\Big(1 + \operatorname{erf}\big(x/\sqrt{2}\big)\Big),$$

where $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\,dt$*.*

3 Remark on densities versus probabilities

A PDF can exceed 1. For example, the uniform PDF on $[0, 1/100]$ is $f(x) = 100$ on its support, yet $\int f = 1$. Density is probability *per unit length*, not probability itself; pointwise probability $P(X = x) = 0$ for any continuous random variable.